CANARA COLLEGE LIBRARY MANGALURU-3

Reg. No.

# BCAAIMS 201

## Second Semester B.C.A. (AI&ML) Degree Examination, June/July 2026 (SEP) (2024-25 Batch Onwards)
## DATA STRUCTURES USING C

Time : 3 Hours | Max. Marks : 80

***Instruction :*** *Answer **any ten** questions from Part – **A** and **one full** question from **each** Unit in Part – **B**.*

PART – A

**(10×2=20)**

1. a) Differentiate primitive data structure and non-primitive data structure.
   b) Name any 4 operations performed by data structure.
   c) Consider the linear array AA(1 : 10). Suppose Base(AA) = 100 and w = 4 words per memory cell for AA. Find the address of AA[5].
   d) Define circularly linked list. Give its diagram.
   e) Differentiate queue and circular queue.
   f) Define dequeue. What are its types ?
   g) Define the terms with respect to tree.
      i) terminal node
      ii) depth of a tree
   h) What is Graph ? Give an example.
   i) Define complete binary tree. Give an example.
   j) Define searching. Name 2 types of searching techniques.
   k) List any two hash functions.
   l) Write one advantage and disadvantage of binary search.

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PART – B (4×15=60)

## Unit – I

2. a) Explain selection and iteration logic in detail with an example.
   b) Explain the representation of two-dimensional array in memory.
   c) Explain Big O notation with an example. **(6+5+4)**

3. a) Explain following :
      a) Graph problem
      b) Combinatorial problems
      c) Geometrical problems.
   b) Explain Algorithm design and analysis process with flow diagram.
   c) What do you mean by traversing a linear array ? Write the algorithm to traverse a linear array. **(6+5+4)**

## Unit – II

4. a) Write an algorithm to PUSH and POP operation on to a stack using linked list.
   b) Write an algorithm to perform insertion operation on queue using arrays.
   c) Evaluate the following postfix expression showing the stack status.
   3, 5, +, 6, 4, –, *, 4, 1, –, 2, ^, + **(5+5+5)**

5. a) Convert the following infix expression into postfix expression using stack status.
   (A + (B * C – (D / E ^ F) * G) * H).
   b) Write an algorithm to perform deletion operation on circular queue using array.
   c) Write an algorithm to insert an item at the beginning of a linked list. **(5+5+5)**

## Unit – III

6. a) With an example explain sequential representation of binary tree.
   b) Write an algorithm for breadth first search(BFS) for a graph.
   c) Draw the binary tree for the following.
   INORDER:E,A,C,K,F,H,D,B,G
   PREORDER:F,A,E,K,C,D,H,G,B **(5+5+5)**

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7. a) Explain Linked Representation of a graph with example.
   b) Draw the binary tree for the expressions given below. Also traverse in preorder, post order and inorder. A * B + (C ^ D/ E/ F).
   c) Write a note on :
      i) Complete binary tree
      ii) Extended binary tree. **(5+5+5)**

## Unit – IV

8. a) Trace the following list of numbers using selection sort technique : 77, 33, 44, 11, 88, 22, 66, 55.
   b) What is linear search ? Write the algorithm for linear search.
   c) Write an algorithm for insertion sort. **(5+5+5)**

9. a) Explain selection sort with an example.
   b) Explain Collision Resolution with Open Addressing.
   c) Write an algorithm for binary search. **(5+5+5)**